

Lógica Computacional 1

Exercícios

Formas Normais para Lógica de Primeira-Ordem

UnB/IE/CIC

Turma 01 - 2024/2

1. Negar e transformar na Forma Normal Clausal:

- (a) $\forall x \forall y (Q(x, y) \rightarrow \exists z (P(x, z) \wedge \exists P(y, z)))$
- (b) $\forall y \neg (\exists x P(x, y) \rightarrow \exists x \neg (P(x, y) \wedge R(y)))$
- (c) $\forall x (P(x) \rightarrow (\forall y (P(y) \rightarrow P(f(x, y)) \wedge \neg \forall y (Q(x, y) \rightarrow P(y)))))$
- (d) $\forall z \forall x \forall y (P(x) \vee \neg P(y) \vee Q(z) \wedge (\neg Q(x) \vee \neg P(y) \vee Q(z)))$
- (e) $\forall x (P(x) \rightarrow Q(x)) \rightarrow (\forall x P(x) \rightarrow \forall x Q(x))$
- (f) $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$
- (g) $\forall x P(x) \rightarrow \forall x Q(x) \wedge \forall x P(f(x)) \rightarrow Q(b)$
- (h) $\forall x \forall y (f(x) \neq f(y) \rightarrow x \neq y)$
- (i) $\forall x \exists y (y = h(x))$
- (j) $(\forall x P(f(x)) \rightarrow \forall x Q(f(x))) \wedge \neg \exists x Q(x) \rightarrow \neg P(f(a))$
- (k) $a = b \rightarrow g(f(a)) = f(g(b))$
- (l) $\forall x P(x) \rightarrow \forall x P(f(x))$
- (m) $\forall x (\neg P(h(h(x))) \rightarrow \neg P(h(x)))$
- (n) $\exists x \exists y (x = f(y) \wedge y = h(x)) \rightarrow \exists x (x = f(h(x)))$
- (o) $P(a) \rightarrow \forall x (\neg P(h(h(x))) \rightarrow \neg P(h(x)))$
- (p) $\forall x P(x) \rightarrow \forall y (R(y, b) \rightarrow P(h(y, b)))$
- (q) $\forall x \forall y (x = y \rightarrow f(x) = f(y))$
- (r) $\forall x \forall y \forall z \forall w (x = y \wedge z = w \rightarrow g(x, z) = g(y, w))$
- (s) $\forall z \exists x \exists y (z = h(x, y) \rightarrow (\forall x \forall y P(h(x, y)) \rightarrow \forall y P(y)))$